Merged Lessons Learned + Deployment Cheat Sheet (Includes Prior PDF Content)

Combines the earlier debugging harvest PDF with the new parameterized deployment template. This supersedes both

1. Critical Lessons Learned (from prior PDF)

```
- Keep environments separate:
  ML training venv / AWS CLI / optional aws-sdk-venv
- model.tar.gz must contain code/inference.py and requirements.txt
- Common failures encountered:
  * /opt/ml/model/code missing -> packaging error
  * numpy.core.multiarray failed to import -> dependency conflict
  * No module named '_loss' -> sklearn version mismatch
- Endpoint creation may take 10–25 minutes to fail because SageMaker retries unhealthy containers.
```

2. Packaging Commands (Prior PDF + Parameterized)

```
PKG_NAME="${MODEL_BASE}_sm_pkg"
MODEL_ARCHIVE="model.tar.gz"

rm -rf "/tmp/${PKG_NAME}"
mkdir -p "/tmp/${PKG_NAME}/code"

cp model/* "/tmp/${PKG_NAME}/"
cp code/*.py "/tmp/${PKG_NAME}/code/"
cp code/requirements.txt "/tmp/${PKG_NAME}/code/"

tar -czf "${MODEL_ARCHIVE}" -C "/tmp/${PKG_NAME}" .
tar -tzf "${MODEL_ARCHIVE}"
```

3. Dependency Pinning Lessons

```
GB model safer pins:
numpy==1.23.5
pandas==1.5.3
scikit-learn==1.2.1
xgboost==1.7.6
joblib==1.3.2

DL models may use torch + newer stacks; do not force GB pins onto DL venv.
```

4. Full Parameter Block (new content)

```
REGION="us-east-2"
ROLE_ARN="arn:aws:iam::219217499832:role/SageMakerExecutionRole"
BUCKET="xt-benefits-model-bucket-fjg"

MODEL_BASE="sdoh-classifier"
VERSION="v1"

PREFIX="${MODEL_BASE}/${VERSION}"
MODEL_NAME="${MODEL_BASE}-${VERSION}"
ENDPOINT_CONFIG_NAME="${MODEL_BASE}-config"
ENDPOINT_NAME="${MODEL_BASE}-endpoint"
MODEL_URI="s3://${BUCKET}/${PREFIX}/model.tar.gz"
```

```
IMAGE_ACCOUNT="763104351884"
IMAGE_REPO="pytorch-inference"
IMAGE_TAG="2.0.1-cpu-py310-ubuntu20.04-sagemaker"
IMAGE="${IMAGE_ACCOUNT}.dkr.ecr.${REGION}.amazonaws.com/${IMAGE_REPO}:${IMAGE_TAG}"

VARIANT_NAME="AllTraffic"
INSTANCE_COUNT=1
INSTANCE_TYPE="ml.m5.large"
```

5. Create Model

```
aws s3 cp "${MODEL_ARCHIVE}" "${MODEL_URI}" --region "${REGION}"

aws sagemaker create-model  --model-name "${MODEL_NAME}"  --execution-role-arn "${ROLE_ARN}"  --primary-containe
```

6. Create Endpoint Config + Endpoint

```
aws sagemaker create-endpoint-config  --endpoint-config-name "${ENDPOINT_CONFIG_NAME}"  --production-variants "V

aws sagemaker create-endpoint  --endpoint-name "${ENDPOINT_NAME}"  --endpoint-config-name "${ENDPOINT_CONFIG_NAM
```

7. Monitoring / Diagnostics (from prior PDF)

```
aws sagemaker describe-endpoint  --endpoint-name "${ENDPOINT_NAME}"  --query '{Status:EndpointStatus,FailureReas

aws logs describe-log-streams  --log-group-name "/aws/sagemaker/Endpoints/${ENDPOINT_NAME}"  --region "${REGION}
```

8. Delete / Recreate

```
aws sagemaker delete-endpoint --endpoint-name "${ENDPOINT_NAME}" --region "${REGION}"
aws sagemaker delete-endpoint-config --endpoint-config-name "${ENDPOINT_CONFIG_NAME}" --region "${REGION}"
aws sagemaker delete-model --model-name "${MODEL_NAME}" --region "${REGION}"
```

9. Invoke Endpoint (Prior PDF content restored)

```
aws sagemaker-runtime invoke-endpoint  --endpoint-name "${ENDPOINT_NAME}"  --content-type application/json  --bo
```

10. Common Failure Signatures (restored + expanded)

```
- zsh command not found: # -> remove comments from pasted blocks
- ProductionVariants length 0 -> broken line continuation
- numpy.core.multiarray failed to import -> dependency conflict
- No module named '_loss' -> training/serving mismatch
- Could not access model data -> role or S3 issue
- Local success can still fail in SageMaker due to container mismatch
```

11. Switching Models

```
Change only:
MODEL_BASE="sdoh-classifier"
or
MODEL_BASE="dl-benefits-classifier"
```

Everything else derives automatically.